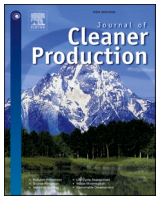
Journal of Cleaner Production 501 (2025) 145316



Contents lists available at ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro





Learning-guided bi-objective evolutionary optimization for green municipal waste collection vehicle routing

Shubing Liao [a,b], Yixin Xu [b], Yunyun Niu [a,*], Zhiguang Cao [b]

[a] School of Information Engineering, China University of Geoscience, Beijing, 100083, China
[b] School of Computing and Information Systems, Singapore Management University, 178902, Singapore

ARTICLE INFO

Handling Editor: Cecilia Maria Villas Bôas de Almeida

Keywords:
Municipal waste collection
Vehicle routing problem
Multi-objective evolutionary algorithm
1DCNN
Nighttime light
NPP-VIIRS

ABSTRACT

Waste management has emerged as a critical issue in modern society, where vehicles are scheduled to visit multiple locations for waste collection and transport. This study focuses on a key problem in waste management: route optimization of waste collection vehicles, and formulate it as a bi-objective vehicle routing problem with stochastic demand (VRPSD), aiming to minimizing both total costs and carbon emissions. Although previous studies have significantly advanced our understanding of solving similar problems, the lack of real-world data and limited problem-solving capabilities still restrict the practical applicability of existing methods. To bridge this research gap, this study designed a regression model using nighttime light data to efficiently and accurately generate two real-case instances in Beijing. Furthermore, a multi-objective evolutionary algorithm integrates Efficient Non-dominated Sorting with Sequential Search and a one-dimensional convolutional neural network (MEAE1C) is proposed to solve the VRPSD problem. MEAE1C integrates a CNN evolver to leverage knowledge from current high-quality solutions to guide subsequent population evolution. Experimental results confirm the superior accuracy in estimates of waste generation, and extensive simulations on benchmark datasets and real-case scenarios consistently demonstrate the superiority of MEAE1C over existing methods. The above results highlight the practical feasibility of the proposed methods in addressing real-world municipal waste management challenges.

1. Introduction

According to the United Nations, nearly 70 % of the world's population will reside in urban areas by the middle of this century (Pons-Prats et al., 2022). The majority of urban population increase will occur in developing countries across Asia and Africa, where urban infrastructure remains comparatively underdeveloped (United NationsH, 2022). The rapid expansion of urban areas and population growth pose challenges for urban systems in providing high-quality urban services and ensuring residents' quality of life, thereby significantly impacting cities' sustainable development. A critical consideration for effective urban systems is how to strategically plan urban services to minimize resource wastage and shorten service time, thereby enhancing the efficiency of the entire urban infrastructure (Korucu and Kucukoglu, 2024).

This study focuses on a crucial urban service: waste collection. In this service, vehicles are scheduled to start from a disposal center, visit a sequence of locations to collect and transport waste. In existing research, this process is often modeled as a Vehicle Routing Problem (VRP), or Location Routing Problem (LRP) if considering location selecting, see Fig. 1. Differ from the application scenes, the basic model can be extended to many variations. For example, to Capacitated VRP (CVRP) when considering the capacity of collection vehicles and to VRP with Time Window (VRPTW) when considering the waste collection time window. There is a special extension for VRP: the VRP with stochastic demand (VRPSD), in which the customer demand is not an accurate value, but obeys a certain distribution. The VRPSD is widely used in this field because the stochastic demand is appeal to the fluctuation of waste production in real-life. Methods for solving the vehicle routing problem with stochastic demand (VRPSD) primarily fall into two categories: exact methods and heuristic methods. Exact methods are characterized by their simple structure and fast problem-solving speeds; however, they often lack the computational power required for large-scale problem solving (Jiang et al., 2022). Heuristic algorithms have better ability to handle large-scale computational problems under various constraints,

This article is part of a special issue entitled: Circular Economy published in Journal of Cleaner Production.

* Corresponding author.
E-mail address: yniu@cugb.edu.cn (Y. Niu).

https://doi.org/10.1016/j.jclepro.2025.145316
Received 11 June 2024; Received in revised form 11 March 2025; Accepted 14 March 2025
Available online 15 March 2025
0959-6526/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

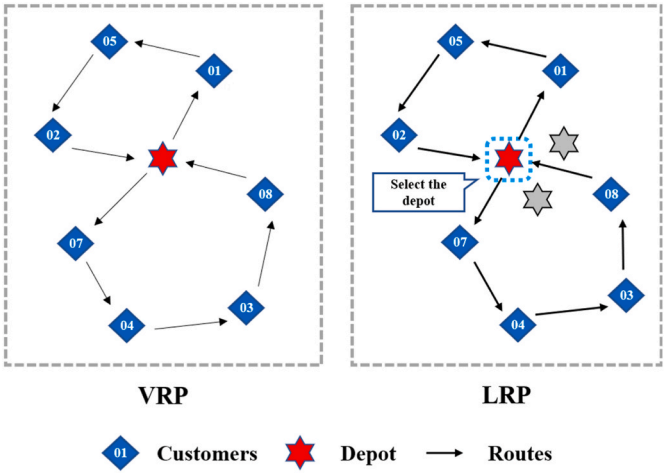


Fig. 1. Example of VRP and LRP

which makes them very popular in solving complex problems (Liao et al., 2024b; Niu et al., 2021a). In real life, VRPSD problems often involve multiple objectives that tend to conflict with each other. Improving one objective often leads to the deterioration of another, making it impossible to optimize all objectives simultaneously (Niu et al., 2023). Some scholars have attempted to convert multiple objectives into a single objective using methods such as weighting, step-by-step constraint, or hierarchical analysis (Wen et al., 2022). Other scholars have adopted the concept of multi-objective optimization to balance multiple objectives by developing multi-objective evolutionary algorithms (Niu et al., 2021b, 2022; Liao et al., 2024a). For example, Niu et al. (2024) proposed a multi-objective optimization algorithm based on a decision tree to solve the location-routing problem in municipal waste management, effectively balancing three objectives by using the decision tree classifier to guide the evolution process.

Since route optimization for waste collection vehicles is a practical problem modeled on real-world scenarios, real-world cases are often used to test the effectiveness of the methods in real-world applications. As illustrated in Table 1, scholars had tried to use many estimation methods to generate real-world cases. However, the methods employed in these studies are relatively simple, primarily relying on statistical data such as Gross Domestic Product (GDP), income levels, population size, and urbanization rates (Kaza et al., 2018; Jassim et al., 2022; Maalouf and Mavropoulos, 2023), or empirical values to estimate waste generation in selected areas. However, waste generation within urban spaces exhibits spatial heterogeneity and is influenced by the above socio-economic factors, which makes the accuracy of these estimation methods is relatively low in community-scale estimation (Lebreton and Andrady, 2019; Li et al., 2022; Zambrano-Monserrate et al., 2021). Therefore, there is an urgent need to seek new data source to present the spatial characteristic of waste production and develop efficient and accurate methods to generate appropriate real-case data. Additionally, it is crucial to fully consider the uncertainty of waste collection amounts inn modeling and optimization for real-world problems. However, considering these uncertainties in modeling and optimization complicates the problem model and significantly increases the time cost when searching for viable solutions (Van England et al., 2020). Therefore, it is necessary to design better heuristics to solve the problem model more effectively.

This study investigates a green municipal domestic waste collection problem for two typical study areas in Beijing. The problem is modeled as a vehicle routing problem with stochastic demand (VRPSD), where stochastic demand is used to represent the fluctuations of customer demand observed in real-life. The contribution of this study lies twofold. First, to address the issue of insufficient real-world data, this study proposes a simple and efficient method to generate real-world cases based on a using a novel data source, nighttime light data. Nighttime light data, collected through remote sensing techniques, represents the brightness of Earth's surface at night. This spatial data is frequently used as a proxy for economic activity in the spatialization of GDP, population, and energy consumption in multiple scales (Zheng et al., 2023) but is rarely used in waste estimation. To achieve more accurate estimation, the proposed method utilizes a Gradient Boosting Regression Tree (GBRT) model to integrate various socio-economic factors into the analysis. The proposed method relies solely on publicly available data, significantly reducing the technical and economic resources required compared to traditional field sampling surveys and sensor-based monitoring approaches (Singh et al., 2023). Its simplicity enables easy application to waste generation estimation in other areas worldwide. Additionally, the high global coverage and high spatial resolution of nighttime light data allow our proposed method to be applied not only to community-scale studies but also to larger scales, such as urban agglomeration studies. Such universal applicability has great potential to alleviate the data sparsity problem globally and facilitate more comprehensive analysis by future researchers. The second contribution of this study lies in introducing a novel algorithm called MEAE1C to solve the VRPSD problem. MEAE1C is a multi-objective evolutionary algorithm that integrates Efficient Non-dominated Sorting with Sequential Search (ENS-SS) (Zhang et al., 2014) and one-dimensional convolutional neural network (1D-CNN) evolver. The MEAE1C algorithm can fully utilize the knowledge of current high-quality solutions to guide subsequent population evolution, thereby discovering more high-quality solutions in the search space. This greatly contrasts with the MOEADT algorithm (Niu et al., 2024), which instantiates the next generation of chromosomes based on specific rules. Simulation results on both Solomon benchmark and real-case data demonstrate that MEAE1C achieves superior optimization values and effectively balances the optimization goals.

Table 1
Comparison between the relevant literature and this study.

Literature	Problem model	Approach	Method to generate real cases
Yazdani et al. (2021)	VRP	Modified hybrid genetic algorithm	Waste generation rate of similar cases in China and Malaysia is used to generate the real cases in Sydney, Australia
Moazzeni et al. (2022)	LRP	Grey wolf optimizer	A hedonic model is used to estimate the demand along 30 different main streets in Edmonton, Canada
Hu et al. (2023)	VRP	Modified metaheuristic algorithm	Statistical data from the urban administrative bureau are used to estimating the average MSW quantities per day of 29 communities in Zhangjiagang, China
Pena et al. (2024)	VRP	Modified genetic algorithm	Conversion factor is used to estimate the waste generation of two cases in New York (USA) and Puerto Real (Spain)
Niu et al. (2024)	LRP	Multi-objective evolutionary algorithm guided by decision tree	Average MSW generation rate obtained from statistical data is used to estimating the waste generation of communities in Fangshan, Beijing, China
This study	VRP	Multi-objective evolutionary algorithm integrates CNN evolver	A GBRT regression model based on spatial data is used to estimate the waste generation of two cases which contains 100 residential areas, respectively, in Beijing, China

The rest of this article is organized as follows. Section 2 introduced the VRPSD model used in this study. Section 3 provided the detail of waste estimation method and the MEAE1D. Experimental results are presented in Section 4 and Section 5 concludes this study.

2. Problem statement

The notations used in this study are shown as follows.

Sets:

N: the customer set, $N = \{n_1, \cdots, n_i\}$.
V: the vehicle set, $V = \{v_1, \cdots, v_i\}$.
Q: the solution set, $Q = \{R_1, \cdots, R_i\}$.
R_i: a solution that contains several routes, $R_i = \{r_1, \cdots, r_j\}$

Parameters

n_i: The residential areas, which are also called the customer nodes in this study
n_0: The waste transfer station, which is also called the depot.
v_i: The vehicles.
x_i, y_i: The horizontal and vertical coordinates of node n_i.
$d_{i,j}$: The Manhattan Distance between node n_i and node n_j.
de_i: The amount of waste needs to be collected in node n_i, which is also called the customer demand of n_i. For n_0, $de_0 = 0$.
t_i^r: The required time to collect waste at node n_i. For n_0, $t_0^r = 0$.
t_0: The required unloading time of v_i at n_0.
r_i: Vehicle v_i's route, which consists of several customer nodes.
c_0: The maximum capacity of vehicles.
C_i^k: The remaining capacity of vehicle v_k after visiting n_i.
T_i: The travel time of v_i in route r_i.
D_i: The travel distance of v_i.

Decision variables:

$P_{i,j}^k$: if vehicle v_k directly travels from n_i to n_j, $P_{kl}^{i,j} = 1$, otherwise $P_{kl}^{i,j} = 0$.

The following assumptions are made in this study.

(1) The coordinates of each node are predetermined and known. The distance between each pair of nodes is measured using the Manhattan distance to better align with reality.
(2) The demand of each node, denoted as de_i, is an independent variable which follows a normal distribution $de_i \sim (\overline{de}_i, \sigma^2)$ with known values of $\overline{de}_i$ and σ.
(3) The amount of waste that needs to be collected at each node, denoted as de_i, can not exceed the maximum capacity of the vehicle, c_0.
(4) A node can only be served by one vehicle at a time. If the amount of waste to be collected at node n_i exceeds the vehicle's remaining capacity, the vehicle will first visit n_i to collect a portion of the waste, then return to the waste transfer station n_0 to unload, and finally revisit n_i to collect the remaining waste.
(5) The fleet is homogeneous. All vehicles have the same maximum capacity c_0, and speed v^0.
(6) The required unloading time, t_0, is the same for all vehicles.
(7) The drivers' normal working hours (denoted as t_n) are constant. If the vehicle's travel time exceeds the driver's working hour, i.e., $T_i > t_n$, overtime wages must be paid.

As shown in Fig. 2, the bi-objective VRPSD in municipal domestic waste collection can be described as follows: vehicle v_k starts from the transfer station n_0, visits all the residential areas to collect the waste, and returns to n_0. Two optimization objectives are considered in this process: minimizing the total cost (including fix cost, oil cost, and drivers' salary) and minimizing the carbon emissions of vehicles. The objectives could be defined as Equation (1):

$$\begin{cases} \text{Min } f_1(x) \\ \text{Min } f_2(x) \end{cases} \tag{1}$$

where $f_1(x)$ and $f_2(x)$ are the total cost and carbon emission, respectively.

Subject to

$$C_i^k - C_j^k = de_j, v_k \in V;\ n_i, n_j \in N \tag{2}$$

$$\sum_{v_k \in V} \sum_{n_i \in N} P_{i,j}^k = 1, n_j \in N \tag{3}$$

$$\sum_{n_i \in N} P_{i,j}^k - \sum_{n_l \in N} P_{j,l}^k = 0, v_k \in V; n_j \in N \tag{4}$$

$$de_j P_{i,j}^k \le \left(C_{j,l}^k - de_l\right) P_{j,l}^k, v_k \in V;\ n_j, n_j, n_l \in N \tag{5}$$

$$P_{i,j}^k = \{0, 1\}, v_k \in V;\ n_j, n_j \in N \tag{6}$$

$$C_i^k > 0, de_i > 0, v_k \in V;\ n_i \in N \tag{7}$$

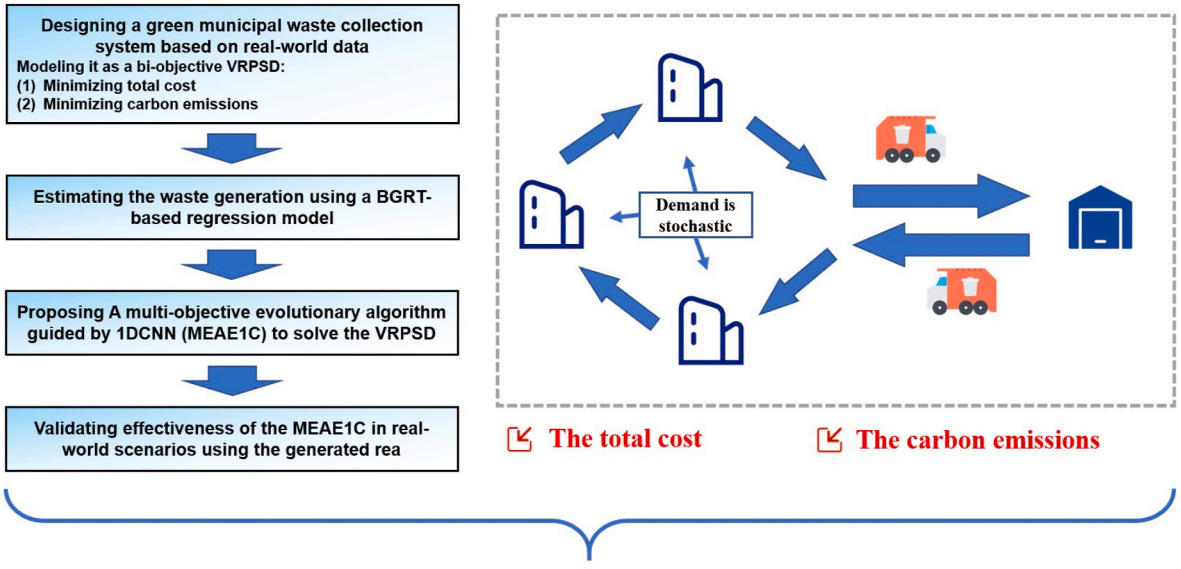


Fig. 2. The brief introduction of this study.

where Equation (2) ensures that after vehicle v_k serves node n_j, the remaining capacity of v_k is reduced by the demand at n_j; Equations (3) and (4) define that vehicle v_k can visit customer n_i only once in a route and that the service for each route is completed continuously; Equation (5) ensures that the remaining capacity of vehicle v_k is sufficient to satisfy the demand of the next node to be visited; Equation (6) defines the binary variable C_i^k and Equation (7) guarantees that both capacity and demand are non-negative.

The total cost, $f_1(x)$, is calculated based on Equation (8):

$$f_1(x) = f_{fix}(x) + f_{oil}(x) + f_{salary}(x) \tag{8}$$

$f_{fix}(x)$ represents the fixed cost of vehicles, which is calculated according to Equation (9):

$$f_{fix}(x) = \alpha \times \mathrm{m} \tag{9}$$

where c is the total cost per vehicle, $\alpha = 500$ CHY; m is the number of vehicles used in a collection plan.

$f_{oil}(x)$ represents oil cost, which is calculated using Equation (10)–(13):

$$f_{oil}(x) = \beta \times \sum_{r_i \in R_j} D_i \tag{10}$$

$$D_i = \sum_{n_k \in r_i} dis(\, n_k) \tag{11}$$

$$dis(\, n_k) = \begin{cases} d_{l,k}, if\ de_k \le C_l^i \\ d_{l,k} + d_{k,0} + d_{0,k}, if\ de_k > C_l^i > 0 \\ d_{l,0} + d_{0,k}, if\ C_l^i = 0 \end{cases} \tag{12}$$

$$d_{l,k} = |x_l - x_k| + |y_l + y_k| \tag{13}$$

where β is the oil cost of vehicle per kilometer, it is an empirical value that represents the average fuel consumption per kilometer of waste collection vehicles under real-world conditions. In this study, $\beta = 10$ CHY.

$f_{salary}(x)$ represents the driver's salary, which is calculated according to Equation (14)–(17):

$$f_{salary}(x) = \sum_{r_i \in R_j} f_s(r_i) \tag{14}$$

$$f_s(r_i) = \begin{cases} \gamma_1 \times\ T_i, \text{if}\ \ T_i \le t_n \\ \gamma_1 \times t_n + \gamma_2 \times (\ T_i - t_n), \text{if}\ \ T_i\ >\ t_n \end{cases} \tag{15}$$

$$T_i = = \sum_{n_k \in r_i} T(\, n_k) \tag{16}$$

$$T(\, n_k) = \begin{cases} \dfrac{dis(\, n_k),}{v^0} + t_k^i, \text{if}\ de_j \le C_i^k \\ \dfrac{dis(\, n_k),}{v^0} + t_k^i + t_0, \text{if}\ de_j > C_i^k \end{cases} \tag{17}$$

where t_n are the normal working hours for drivers, $t_n = 8$; γ_1 is the regular salary for drivers, $\gamma_1 = 10$ CNY per hour; γ_2 is the overtime salary for drivers, $\gamma_2 = 20$ CNY per hour.

In today 's green economy, minimizing the carbon emissions is a very important goal in municipal domestic waste collection system. Therefore, minimizing the amount of carbon emissions is also considered as an objective in this study, denoted as $f_2(x)$. Referring to previous studies (Niu et al., 2024; Liao et al., 2024a), the calculation of carbon emission cost is shown in Equation (18):

$$f_2(x) = \delta \times \sum_{r_i \in R_j} D_i \tag{18}$$

where δ is the carbon emission of the vehicle per kilometer, $\delta = 0.25$ kg.

3. Method

3.1. Estimating of municipal domestic waste generation

3.1.1. Study area and research data

This study chooses Beijing as the study area. Beijing, the capital of China, is one of China's mega-cities. It comprises 16 districts and counties, with an annual permanent population exceeding 10 million. As reported by the Beijing Environmental Sanitation Group (BESG), the amount of municipal domestic waste collected by BESG was 7.4 million tons in 2022. Two typical areas are investigated as case studies in this study: one involves municipal domestic waste collection in the boundary of three urban core districts, and the other involves municipal domestic waste collection in the urban-rural boundary. As shown in Fig. 3, each case contains a transfer station and 100 residential areas.

This study constructs a spatial model to estimate household waste generation based on nighttime light data and socio-economic statistics of the study areas. The nighttime light data used in this study is the Annual VNL V1 global nighttime light data product with a spatial resolution of 500 m (Elvidge et al., 2021), which is provided by the Earth Observation Group, Payne Institute for Public Policy, and Colorado School of Mines (https://eogdata.mines.edu/products/vnl/). Socio-economic data is sourced from the "Beijing Statistical Year-book" (https://tjj.beijing.gov.cn/). Since the earliest NPP-VIIRS data was released in 2012, the study period in this paper spans from 2012 to 2022. Based on findings from previous studies mentioned in Section 1 and offline correlation tests, the socio-economic indicators with Pearson correlation coefficient values greater than 0.4 will be selected in this study, which include permanent population, per capita disposable income, per capita consumption expenditure, total GDP, and per capita GDP. The sum of nighttime light intensity is selected as the indicator of nighttime light. The boundary of each residential area is obtained using QGIS software based on OpenStreetMap (https://www.openstreetmap.org).

3.1.2. Calculation of municipal domestic waste amount

The estimation of the amount of municipal domestic waste generated by each residential area can be divided into two steps. Firstly, a data spatialization method is established to convert large-scale statistical waste generation data into spatial data, enabling it to be analyzed and visualized in micro scale. There are some similar works studying on the spatialization of GDP (Del Castillo et al., 2024) and population density (Wang et al., 2018). Referring to their works, the functional relationship between independent variables (indicators) and the dependent variable (waste generation amount) is modeled using several algorithms: Gradient Boosting Regression Tree (GBRT) Bayesian Ridge Regression (BRR), Multiple Linear Regression (MLR), Elastic Net Regression (ENR), Support Vector Regression (SVR) and Random Forest Regression (RF) according to Equation (19):

$$y = f(x_1, x_2, \ldots, x_n) \tag{19}$$

where y is amount of municipal domestic waste (ton per year), x_1 to x_n are the indicators.

Three error measurements are used to evaluate the accuracy of the established regression models: Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE) and R-squared (R^2). The regression accuracy of GBRT, BRR, MLR, ENR, SVR are compared and the model with the highest accuracy will be selected for the actual estimation. In this process, it is assumed that socio-economic indicators remain consistent across residential areas within the same district, thereby not influencing the amount of waste generation in different residential areas. Based on this assumption, the amount of waste generation in different residential area is only affected by the other indicator:

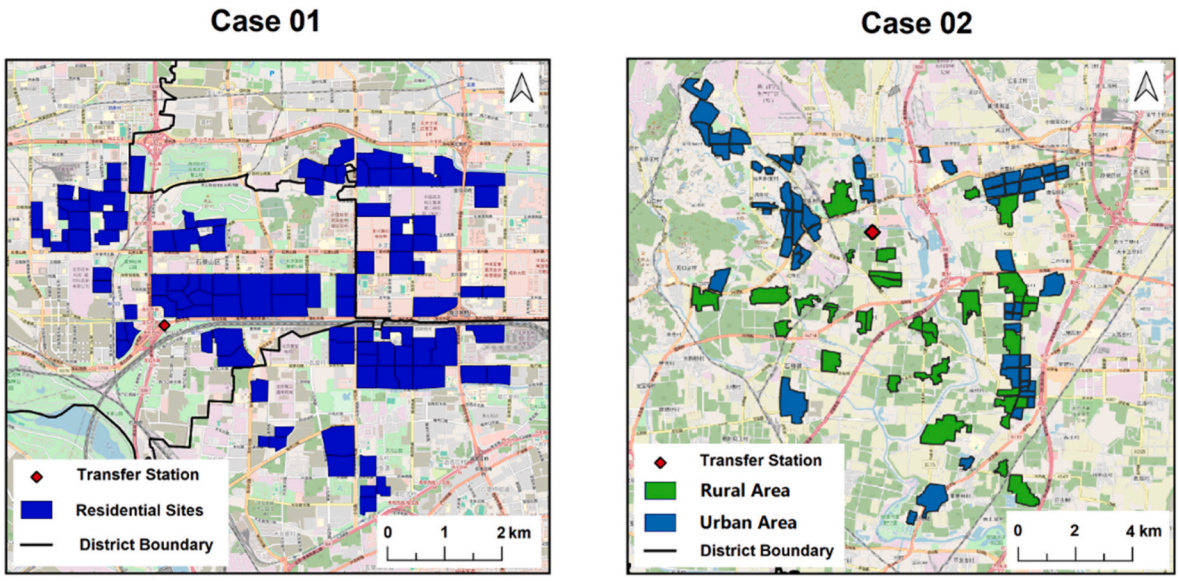


Fig. 3. Illustration of the selected cases in Beijing.

nighttime light intensity. The waste generation can be calculated using Equation (20):

$$y(n_i) = f(x_1(n_i), x_2(n_i), \ldots, x_{n-1}(n_i), NTL(n_i)) \quad (20)$$

where $y(n_i)$ is the amount of municipal domestic waste generated by node n_i; $x_1(n_i)$ to $x_{n-1}(n_i)$ are the socio-economic indicators of the district where n_i node is located; $NTL(n_i)$ is the sum of nighttime light intensity of node n_i.

The sum of nighttime light intensity for node n_i, $NTL(n_i)$, is extracted using spatial analysis methods in QGIS software. Nighttime light intensity is commonly depicted as an image consisting of pixels. However, not all pixels overlap with the residential zone, and some pixels only partially intersect with it. Only the pixels covering a sufficient area within the residential zone should be counted in the calculation. To determine which pixels should be counted, this study divide each pixel into 4 equal-sized windows. Only those windows with more than half of their area overlapping with the residential zone are considered effectively covered and counted in the calculation, see Fig. 4.

3.2. The MEAE1C algorithm

In this study, a multi-objective evolutionary algorithm, named MEAE1C, is proposed to solve the waste collection problem. MEAE1C integrates Efficient Non-dominated Sorting with Sequential Search (ENS-SS) and a one-dimensional convolutional neural network (1D-CNN) evolver. The framework of MEAE1C is illustrated in Algorithm 1 and Fig. 5. MEAE1C first generates the initial population and then uses ENS-SS to sorts solutions within a population by their qualities. Afterwards, a CNN evolver is employed to guide the population evolution process and generate a new population. Distinct from the MOEA-DT algorithm (Niu et al., 2024), which directs population evolution by generating high-quality populations using a certain number of rules learned by decision tree classifier, the CNN evolver in MEAE1C leverages knowledge from the current high-quality solutions to evaluate solutions generated by the stochastic Push Forward Insertion Heuristic (PFIH) (Niu et al., 2022). This approach allows the MEAE1C fully use the knowledge of current high-quality solutions to lead subsequent population evolution, thereby to discover more high-quality solutions in the

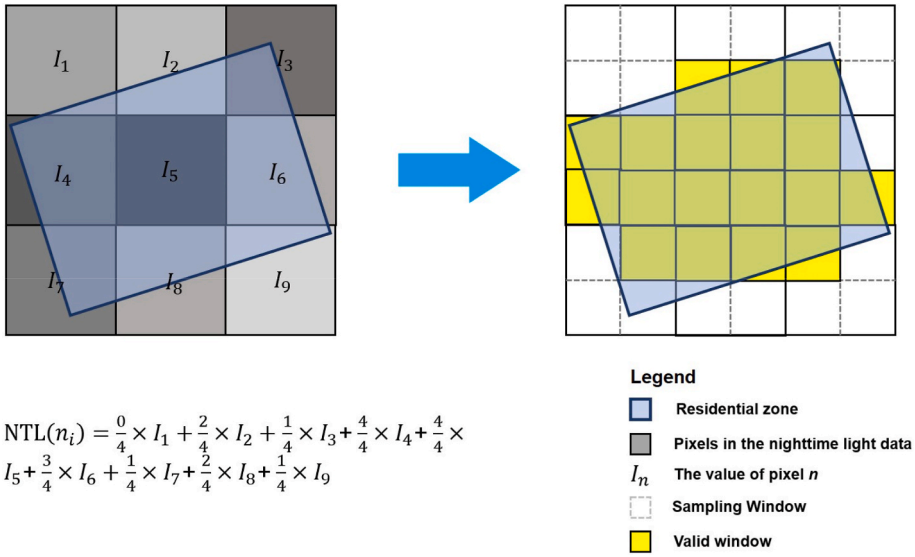


Fig. 4. Extraction of $NTL(n_i)$ for residential area n_i.

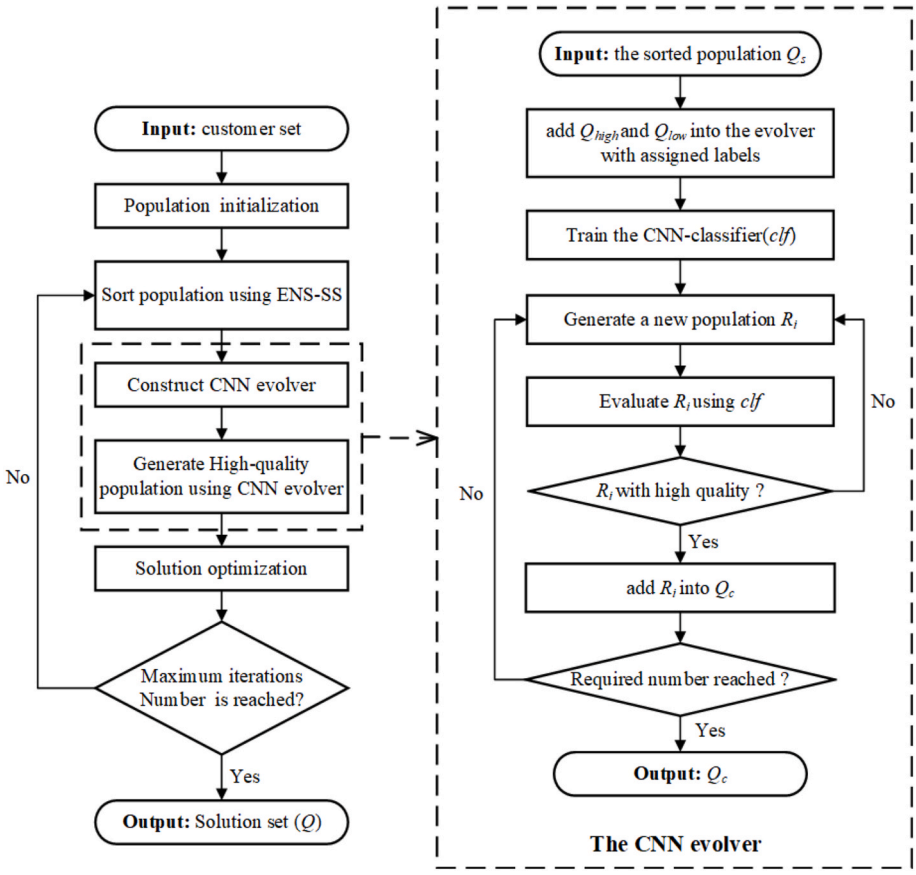


Fig. 5. Flowchart of the MEAE1C.

search space, which greatly contrasts with the instantiation of the next generation of chromosomes based on specific rules learned by a machine learning algorithm. This sorting-evolution process is iterated multiple times until reaching the maximum number of iterations, at which point the population evolution process terminates and the solution set Q is returned.

Algorithm 1. The MEAE1C Algorithm

<!–Col Count:1–>**Input:** N' (the data set), size (population size)
Output: Q
1: $P \leftarrow$ initialization (N', size)
2: $P_c \leftarrow$ sorting (P)
3: $Q \leftarrow$ create empty set
4: evolver $\leftarrow$ create CNN evolver
5: **while** maximum iteration number is not reached do
6: $Q \leftarrow$ evolution (P_c, evolver)
7: $Q_c \leftarrow$ crossover
8: $Q_c \leftarrow$ mutation
9: **end while**
10: $Q \leftarrow Q_c$
11: **return** Q

3.2.1. Chromosome representation

In evolutionary algorithms, solutions are often encoded as chromosomes to map instances to the search space. Each element in a chromosome corresponds to a customer (or residential area), with its value representing its priority in the routes. Customer with higher priority values is positioned earlier in the routes.

For example, the chromosome {6,3,8,1,2,7,5} can be translated as {6(n_1), 3(n_2), 8(n_3), 1(n_4), 2(n_5), 7(n_6), 5(n_7)}. This illustrates that the first customer n_1 holds a priority of 6, and the second customer n_2 holds a priority of 3. Note that priority value 4 is missing in the example chromosome as it marks the separation between two routes. When decoding this chromosome into routes, the transfer station n_0 is selected as the start node of each route and customers are added progressively in descending order of their priority values. Upon encountering a missing priority value, the previous route is concluded by appending customer n_0 as the end node and the generation of a new route will start. In the example chromosome, the priority of customers can be ordered as n_3, n_6, n_1, n_7, _, n_2, n_5, n_4, with the blank indicating the separation between two routes. Following this priority, the first route of can be decoded as $r_1 = \{n_0 \rightarrow n_3 \rightarrow n_6 \rightarrow n_1 \rightarrow n_7 \rightarrow n_0\}$, and the second route can be decoded as $r_2 = \{n_0 \rightarrow n_2 \rightarrow n_5 \rightarrow n_4 \rightarrow n_0\}$.

By assigning each customer a unique priority value, a one-to-one correspondence between the chromosome and the solution is ensured. This approach enables the direct use of existing chromosomes in subsequent population evolution and mutation processes, eliminating the need to repeatedly modify the mapping rules between the chromosome and the solution.

3.2.2. Population initialization

The first step of evolutionary algorithms is to determine an initial solution, i.e., the initial routes for vehicles. All vehicles should start with and end at the transfer center n_0. Therefore, n_0 is the starting and ending node for all routes. To enhance the validity of the solution set, the initial solution is generated by gradually adding customers in the order of their IDs, with each customer's demand set to the average demand, de_i. The following criteria are adhered to when generating routes in the initial solution (Niu et al., 2024):

1. The sum of customer demand in any route must not exceed the maximum capacity of the collection vehicles, c_0.

2. The service time of each vehicle must not exceed the allotted time limit.

Once all customers are included in the initial solution, the number of routes in this solution is determined. This number is set as the maximum number that the number of routes in all solutions within the solution set can reach. Subsequently, customers are randomly added to form new solutions until the required number of solutions is generated. This solution set will serve as the initial population for subsequent operations.

3.2.3. Population evolution guided by CNN

After generating the initial population, a Pareto sorting operation is per-formed using ENS-SS, which will be used to create the next generation population. The new population consists of two parts: one part is newly generated using the CNN evolver; and the other part is obtained by optimizing the top 50 % of chromosomes from the previous population. Each part comprises 50 % of the new population. The evolution of the population continues until the maximum number of iterations is reached.

The one-dimensional convolutional neural network (1D-CNN) is primarily used for processing one-dimensional data, such as time series data, text data, and signal processing (Kiranyaz et al., 2021). The typical 1D-CNN structure is shown in Fig. 6. Compared to other machine learning algorithms such as decision tree classifiers, 1D-CNNs are more effective in capturing local patterns and features within the input data.

In a population, if a chromosome is classified in the upper echelon following Pareto sorting, it indicates that this chromosome is superior to many other chromosomes in the population. After Pareto sorting, a certain proportion (H %) of chromosomes in the upper part of the current population are regarded as "high-quality" solutions (Q_{high}), and chromosomes in the lower part of the population (L %) are considered as "low-quality" populations (Q_{low}). These sorting results are further used to train a CNN evolver to classify whether a newly generated chromosome belongs to (Q_{high}) or (Q_{low}). A stochastic push forward insertion heuristic (PFIH) (Niu et al., 2022) is applied to generate new chromosomes. Each newly generated chromosome is evaluated by the evolver to predict whether it is high-quality or low-quality. Only those recognized as high-quality chromosomes will be stored. When the number of stored high-quality chromosomes meets the required number, all stored chromosome will be added to the next generation population. The main procedure of the CNN evolver is shown in Fig. 5.

3.2.4. Solution optimization

In section 3.2.3, it is mentioned that the top 50 % of the current population chromosomes will be optimized and used in the next generation. As illustrated in Fig. 7, this strategy encompasses three operators: the route crossover operator, the short-route merging operator, and the long-route split operator. The route crossover operator randomly selects two different routes within a solution and exchanges two arbitrarily intercepted segments from these routes. The short-route merging operator combines two short vehicle routes into one, while the long-route split operator divides a long route into two separate routes. Once the current population is determined, the objective values of each chromosome are calculated using the Route Simulation Method (RSM) (Niu et al., 2024). To ensure generality, the RSM runs 10 times and compute the average objective values. Subsequently, ENS-SS is utilized to establish the Pareto dominance relationships among chromosomes in the population. If the current population includes superior individuals, the archive set Q is updated and these individuals will be stored in set Q. A sorting algorithm is then applied to sort Q and the current population, with the aim of identifying the elite individuals. The size of Q is fixed and is set to match the size of the population.

4. Results

4.1. Data and settings

This section presents simulation results for the performance of the proposed MEAE1C algorithm. Six instances from the Solomon benchmark are modified to serve as the dataset for MO-VRPSD. BESG collects municipal waste twice a day in each residential area, and the proportion of dry waste to wet waste generated by residents is approximately 2:1 (Liu et al., 2024). Therefore, the average demand for each node is set as one-third of the value calculated by the regression model; the standard deviation is randomly set to be 1/5 to 1/3 of the average demand; The coordinates of the center of each residential area are assigned as the coordinates of the corresponding node. Consequently, the datasets for the two case studies are generated and named BJ01 (Case 1) and BJ02 (Case 2), respectively. The vehicle capacity for these two case studies is set to 2100 kg and the normal work hours is set as 2 h based on our survey of waste collection at household level in Beijing.

The parameters and interpretations of the experiments are determined based on offline testing and previous studies (Niu et al., 2021a, 2024). The architecture of the 1D-CNN includes five convolutional

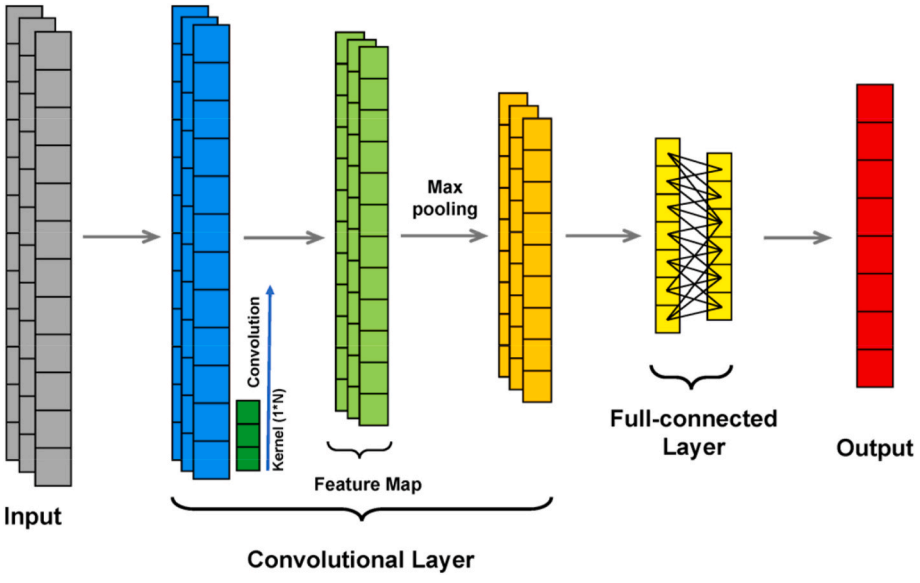


Fig. 6. The typical 1DCNN structure.

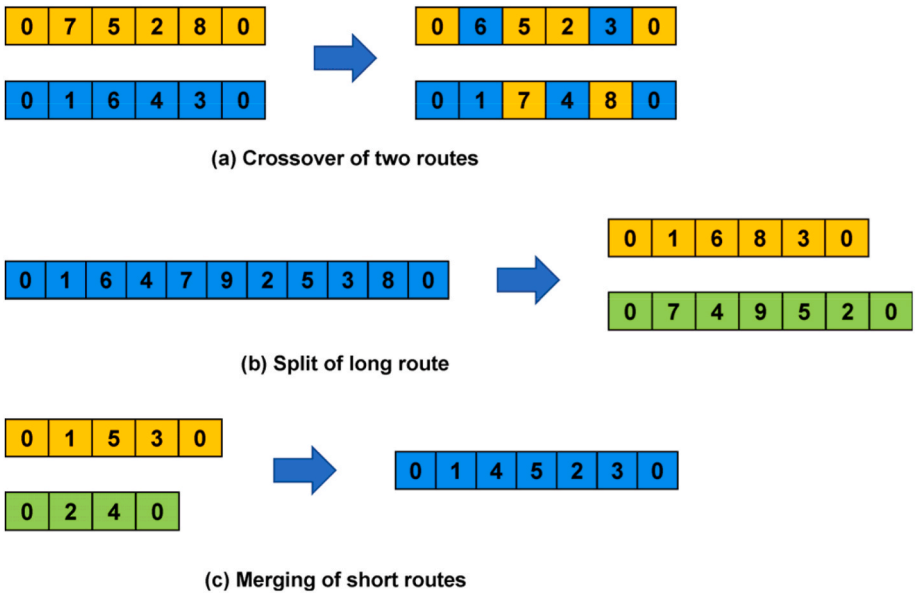


Fig. 7. The solution optimization strategy.

layers with a kernel size of 3. The output from the final convolutional layer is flattened and passed through two fully connected layers. The first fully connected layer uses ReLU activation, and the second is the output layer with 2 labels for classification. The model is trained using the Adam optimizer and the loss function is cross-entropy loss.

Two objectives, total cost and carbon emissions, are used to evaluate the performance of the proposed algorithm. Two indicators, Inverted Generational Distance (IGD) and *Spacing*, are employed to evaluate the quality of the solutions. IGD quantifies the average Euclidean distance between all solutions in the reference Pareto front and the non-dominated solutions. A lower IGD value indicates that the non-dominated solution set is closer to the reference Pareto front and more uniformly distributed, thereby signifying better convergence and diversity. Spacing, on the other hand, measures the distribution of the population. A smaller Spacing value signifies a more evenly distributed population, reflecting a better solution distribution.

4.2. Accuracy of the regression model

The accuracy of the regression models is evaluated through a ten-fold cross-validation method and illustrate results in Table 2 and Fig. 8. The RSE value of the GBRT-based model is 4.98, which is lower than those obtained using the MLP (7.28), BRR (6.36), ENR (6.39), SVR (6.18) and RFR (5.02). The regression model using GBRT achieves 8.88 % and 0.93 for MAPE and R^2, respectively, outperforming the MLP (14.44 % and 0.81), BRR (12.23 % and 0.84), ENR (12.28 % and 0.85), SVR (11.23 % and 0.87) and RFR (8.96 % and 0.91). These results indicate that the GBRT-based regression model demonstrates superior accuracy in modeling the generation of municipal domestic waste compared to the other five methods, corroborating the findings of previous studies (Wan et al., 2024; Lu et al., 2022).

Table 2
Comparison of the accuracy of different regression models.

Method	RSE	MAPE (%)	R^2
MLR	7.28	14.44	0.81
BRR	6.36	12.23	0.84
ENR	6.39	12.28	0.85
SVR	6.18	11.23	0.87
RFR	5.02	8.96	0.91
GBRT	4.98	8.88	0.93

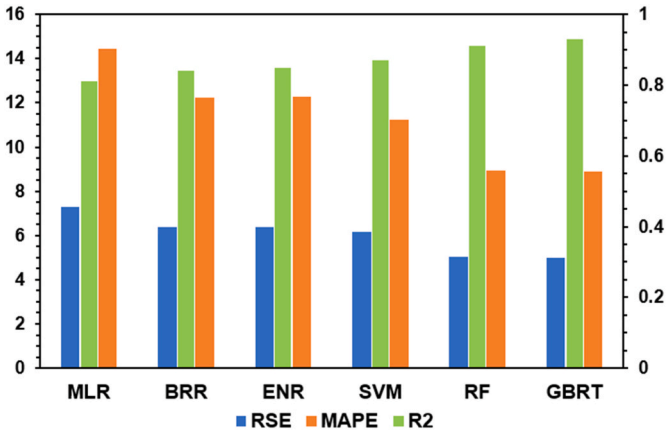


Fig. 8. Comparison of the accuracy of different regression models.

4.3. Effectiveness of the CNN evolver

As mentioned in Section 3.2, the proposed MEAE1C algorithm employs a CNN evolver to learn knowledge from previous high-quality solutions and guide the population evolution. The effectiveness of the CNN evolver is showcased using the simulation results. In particular, the MEAENC is used to denote MEAE1C without the CNN evolver while

Table 3
Effectiveness of the CNN evolver.

Instance	Algorithm	Total cost	Carbon emission	IGD	Spacing
C1	MEAE1C	61529.09	1010.16	403.02	2453.57
	$MEAE_{NC}$	73123.01	1091.84	953.61	5412.63
C2	MEAE1C	39318.28	938.65	170.96	3153.74
	$MEAE_{NC}$	43563.77	956.83	368.13	4253.69
R1	MEAE1C	45148.25	843.79	179.26	3160.93
	$MEAE_{NC}$	59884.60	940.65	265.28	4569.12
R2	MEAE1C	35533.61	765.70	122.33	3472.38
	$MEAE_{NC}$	39773.28	803.25	196.53	4062.33
RC1	MEAE1C	59951.03	1142.48	379.54	5154.50
	$MEAE_{NC}$	65326.89	1205.93	547.66	7764.39
RC2	MEAE1C	40741.68	959.43	183.62	236.93
	$MEAE_{NC}$	46983.26	1002.33	283.35	886.44

keeping all other parameters unchanged. The results are shown in Table 3 and Fig. 9. Compared with MEAENC, the solutions generated by MEAE1C achieve lower average total cost values and lower average carbon emission values. The gap of total cost between the MEAE1C and MEAENC ranges from 10 % to 30 % across the six instances, while the gap of carbon emission ranges from 3 % to 12 %. Moreover, the solution sets generated by MEAE1C exhibit superior quality compared to those generated by MEAENC. The simulation results support that CNN evolver can help the proposed MEAE1C algorithm generate better solutions.

4.4. Comparison with other algorithms

This study compares the proposed MEAE1C algorithm with two traditional methods: the Gurobi and multi-objective ant colony system (MOACS) (Li et al., 2022) in solving two small instances M1 and M2. These two instances are obtained by randomly selecting 10 customers from C1 and R1. Moreover, three state-of-the-art (SOTA) methods in solving similar VRPSD problems: MOEADT (Niu et al., 2024), MOEA (Tan et al., 2007) and MRDL (Gee et al., 2016), are also compared with the MEAE1C in in the modified Solomon benchmark. The average objective values of the optimal solutions generated by these algorithms are shown in Table 4 and Fig. 10. Simulation results show that the proposed MEAE1C algorithm presents superior efficiency in solving bi-objective VRP-SD in municipal waste collection. In small instances, MEAE1C has achieved close total costs and carbon emission values to those of Gorubi and MOACS. Compared with MOEA and MRDL, MEAE1C achieves both low total costs and carbon emission values in the solution sets. The average total cost values of those generated by the MEAE1C are below 70 % of those generated by the MOEA and MRDL on the six instances, and the average carbon emission values of those generated by the MEAE1C are approximately 80 % of those generated by the MOEA and MRDL. Compared with the solutions generated using the MOEADT, the pro-posed MEAE1C algorithm can reduce nearly 10 % of the total cost and achieve a certain decrease in carbon emissions.

4.5. Case study

As mentioned in Section 3.1 and Section 4.2, this study investigates two case studies in Beijing. Here, we use these two case studies to test the efficiency of the proposed MEAE1C algorithm in real-life scenarios, comparing its performance with that of MOEA, MOEADT and MRDL. The results are shown in Table 5 and Fig. 11. The average total cost value and carbon emission value of the solutions generated using the MOEAE1C algorithm are all better than the other three methods. The average total cost values of these generated by the MEAE1C are nearly 70 % of those generated by the MOEA and MRDL in both case of BJ01 and BJ02; the average carbon emission values of these generated by the MEAE1C are approximately 75 % of those generated by the MOEA and MRDL in both case studies. Compared with the solutions generated using the MOEADT, the proposed MEAE1C algorithm can reduce nearly 4 % of the total cost and a decrease of 6 % of carbon emission. The results of two case studies all verify the efficiency of the proposed MEAE1C algorithm.

These two case studies both embody the advantage of estimating waste generation using nighttime light data. It is true that we can obtain waste generation data from residents in these areas through surveys. Additionally, IoT technology can be employed to monitor the real-time capacity of waste bins in these residential areas, thereby providing waste generation data. However, surveys not only require significant human resources, but also necessitate the development of different models based on the specific characteristics of each region, due to variations in income levels and consumption habits (BJ 01). This is crucial to ensure the accuracy of the estimates. Real-time monitoring of waste bin capacity using IoT in conjunction with GPS is currently one of the most accurate methods for obtaining waste generation data in residential

Table 4
Comparison with other algorithms.

Instance	Algorithm	Total cost	Carbon emission	IGD	Spacing
M1	MEAE1C	7263.51	125.36	63.25	453.55
	MOACS	7335.02	125.93	65.32	486.65
	Gurobi	7395.33	126.85	69.26	531.36
M2	MEAE1C	5026.23	92.75	30.36	296.32
	MOACS	5082.36	93.16	33.37	332.53
	Gurobi	5113.48	93.73	36.01	356.42
C1	MEAE1C	61529.09	1010.16	403.02	2453.57
	MOEADT	70123.01	1011.84	630.45	3825.72
	MOEA	83756.66	1228.27	1832.66	7690.04
	MRDL	82204.42	1027.10	1527.12	8652.36
C2	MEAE1C	39318.28	938.65	170.96	3153.74
	MOEADT	41329.01	938.61	179.37	3867.07
	MOEA	51043.73	1017.70	316.81	4475.17
	MRDL	48693.00	943.19	237.20	4102.12
R1	MEAE1C	45148.25	842.79	179.26	3160.93
	MOEADT	53884.33	840.94	248.53	4569.12
	MOEA	75521.60	972.46	1532.47	12890.12
	MRDL	66325.23	906.16	1326.54	8652.36
R2	MEAE1C	35533.61	765.70	122.33	3472.38
	MOEADT	36940.91	789.38	193.04	3774.69
	MOEA	45186.19	925.62	672.46	3886.46
	MRDL	38141.28	940.05	875.82	0.00
RC1	MEAE1C	59951.03	1142.48	379.54	5154.50
	MOEADT	62692.36	1143.97	442.21	7326.28
	MOEA	88825.44	1191.93	872.33	9408.37
	MRDL	71728.04	1173.01	1023.65	21727.51
RC2	MEAE1C	39741.68	959.43	183.62	236.93
	MOEADT	41668.10	982.56	235.73	752.96
	MOEA	57003.38	1061.75	476.27	0.00
	MRDL	49568.96	957.15	523.40	0.00

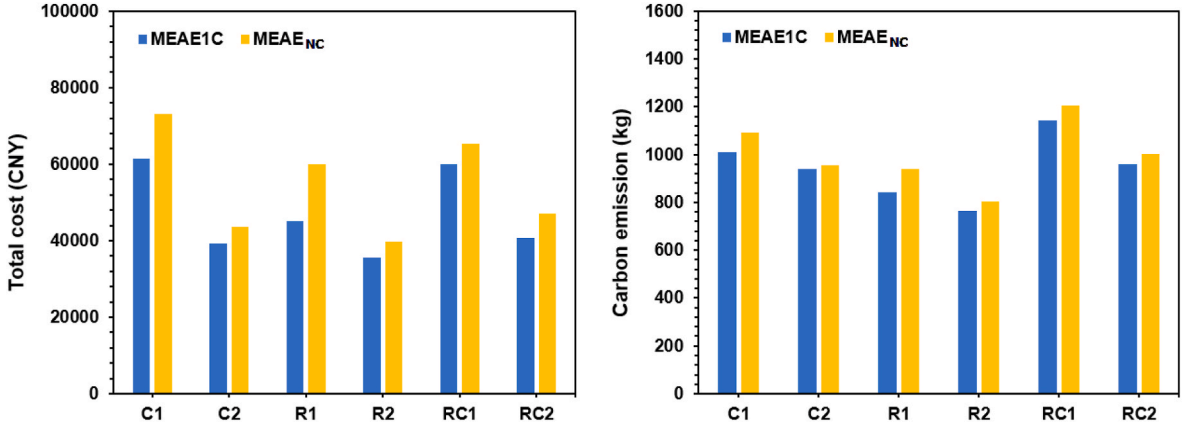


Fig. 9. Comparison of MEAE1C with MEAE$_{NC}$

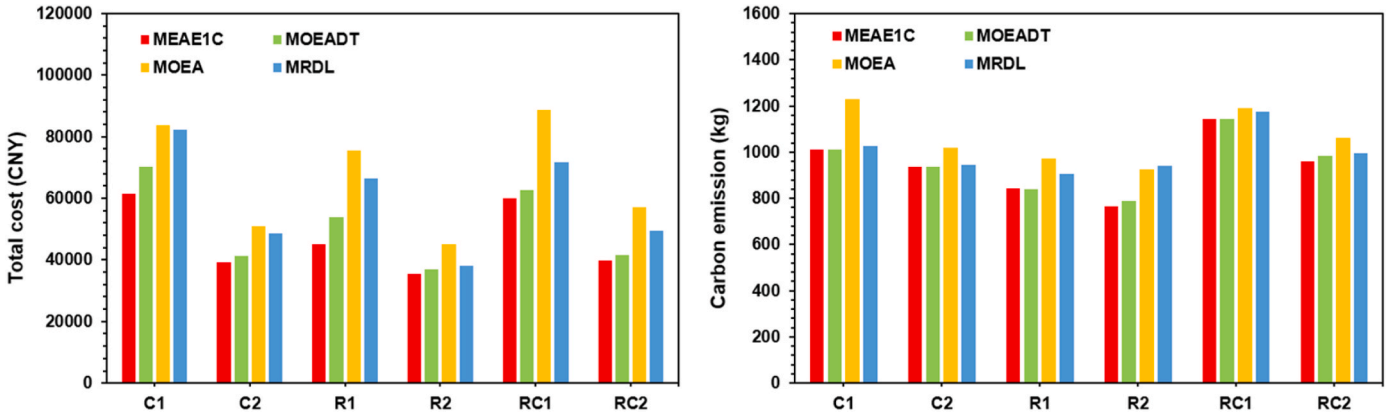


Fig. 10. Comparison of MEAE1C with MOEADT, MOEA and MRDL

Table 5
Result of two case studies.

Instance	Algorithm	Total cost	Carbon emission	IGD	Spacing
BJ01	MEAE1C	6213.32	46.33	136.26	14.15
	MOEADT	6309.35	49.33	140.23	16.02
	MOEA	8523.16	52.06	198.55	28.85
	MRDL	9016.33	53.11	201.36	29.06
BJ02	MEAE1C	13365.12	291.38	235.27	12.07
	MOEADT	13333.64	305.15	272.53	13.56
	MOEA	18253.15	338.52	356.36	31.05
	MRDL	19576.56	349.38	358.73	30.86

areas (Singh et al., 2023). However, applying IoT technology to monitor waste generation in typical urban residential areas is still a significant challenge in developing countries like China. When it comes to rural areas, such as in BJ02, the cost will become even more unacceptable considering the high maintenance costs. The proposed method relies solely on publicly available data, significantly reducing the technical and economic resources required. Additionally, nighttime light data offers several advantages, including a long-time span (since 2021), global coverage, high spatial resolution (500m × 500m), and a short revisit cycle (15 days). Therefore, in future studies, high-precision estimation models can be established to utilize nighttime light data for mapping waste generation at various scales, thereby facilitating the analysis of the spatiotemporal variability of waste production.

5. Conclusion

This study designed a green municipal domestic waste collection system for two typical study areas in Beijing, the capital of China. The system is modeled as a Vehicle Routing Problem with Stochastic Demand (VRPSD) while aiming to minimize both total cost and carbon emissions. To overcome the challenge of insufficient real-world data, this study proposes a GBRT-based regression model, which incorporates social-economic indicators and nighttime light intensity, to estimate the amount of waste generation in residential areas. Furthermore, this study proposes a multi-objective evolutionary algorithm named MEAE1C to solve the VRPSD problem. MEAE1C begins by generating an initial solution and then progressively "evolves" the solutions to improve their quality. During this evolution process, MEAE1C uses ENS-SS to assess the quality of solutions. High-quality of solutions from previous iterations further guide subsequent evolution through a CNN evolver, enabling the discovery of more high-quality solutions in the search space.

Evaluations demonstrate that GBRT-based regression model achieves superior accuracy when modelling the municipal domestic waste generation. The RSE value of the GBRT-based model is 4.72, which is lower than those using other estimation methods such as Multilayer Perceptron (MLP) and Bayesian Ridge Regression (BRR). The MAPE and R2 values of GBRT-based model are 8.47 % and 0.92, respectively, which are also better than those of the other estimation models. The effectiveness of MEAE1C is verified by simulating on six instances of Solomon benchmark and two real-case data generated by GBRT-based model. Comparing with other state-of-the-art methods, MEAE1C yields better optimization values across all datasets, indicating that the proposed algorithm can better balance the total cost and carbon emission. Compared with other methods, MEAE1C reduces the average total cost values by up to 40 % and the average carbon emission values by up to 20 %.

Although the results proved the designed data spatialization method

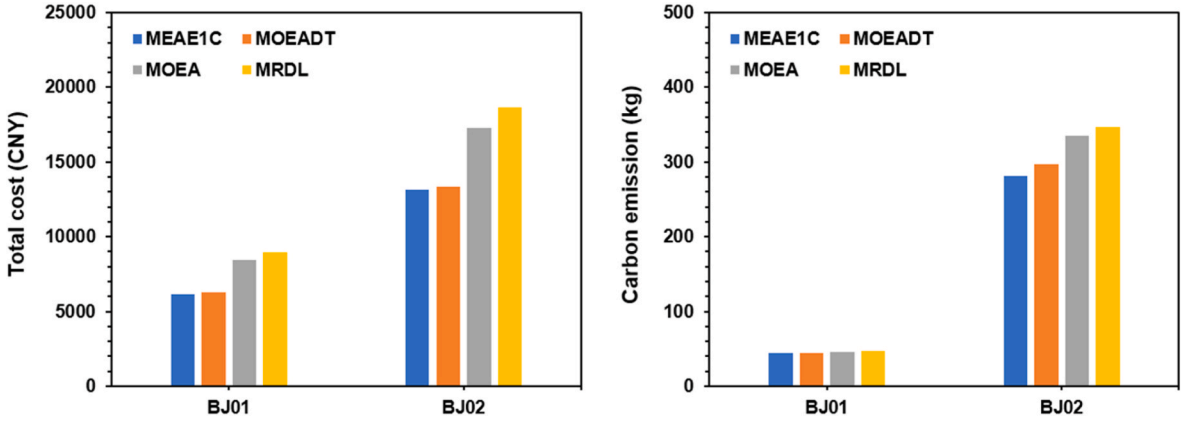


Fig. 11. Results of the two case studies in Beijing.

and the proposed MEAE1C algorithm can efficiently address the issue of insufficient real-world data and problem-solving capability of existing methods in municipal waste collection, there are still some shortcomings that cannot ignored. Firstly, this study considers only a subset of factors influencing the annual variation in municipal waste generation in the study area when establishing the waste generation regression model. In reality, waste generation exhibits not only inter-annual variations but also seasonal fluctuations. Additionally, due to differences in consumption levels and consumption patterns, the composition of municipal waste varies across different residential areas. Therefore, further analysis of these factors is required to more accurately estimate municipal waste generation. Secondly, in the VRP modeling, for computational convenience, this study did not consider the impact of vehicle load changes on total cost and carbon emissions during travel; nor did this study account for potential future changes in municipal waste generation when generating actual data. Further studies should comprehensively consider these factors to establish a problem model that more accurately reflects real-life scenarios.

CRediT authorship contribution statement

Shubing Liao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Yixin Xu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software. **Yunyun Niu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization. **Zhiguang Cao:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China [grant numbers 62172373, 61872325]; the Fundamental Research Funds for the Central Universities, China [grant number 2652019028] and China Scholarship Council. The authors gratefully acknowledge the constructive suggestions and comments on this paper from the editors and anonymous reviewers.

Data availability

Data will be made available on request.

References

Del Castillo, M.F.P., Fujimi, T., Tatano, H., 2024. Estimating sectoral covid-19 economic losses in the Philippines using nighttime light and electricity consumption data. Front. Public Health 12, 1308301. https://doi.org/10.3389/fpubh.2024.1308301.

Elvidge, C.D., Zhizhin, M., Ghosh, T., Hsu, F.C., Taneja, J., 2021. Annual time series of global VIIRS nighttime lights derived from monthly averages: 2012 to 2019. Remote Sens. 13, 922. https://doi.org/10.3390/rs13050922.

Gee, S.B., Arokiasami, W.A., Jiang, J., Tan, K.C., 2016. Decomposition- based multi-objective evolutionary algorithm for vehicle routing problem with stochastic demands. Soft Comput. 20, 3443–3453. https://doi.org/10.1007/s00500-015-1830-2.

Hu, S., An, L., Shen, L., 2023. A multi-objective modeling and optimization approach to municipal solid waste collection for classified treatment in China towards sustainable development. Sustain. Cities Soc. 98, 104846. https://doi.org/10.1016/j.scs.2023.104846.

Jassim, M.S., Coskuner, G., Zontul, M., 2022. Comparative performance analysis of support vector regression and artificial neural network for prediction of municipal solid waste generation. Waste Manag. Res. 40, 195–204. https://doi.org/10.1177/0734242X211008526.

Jiang, Y., Wu, Y., Cao, Z., Zhang, J., 2022. Learning to solve routing problems via distributionally robust optimization. In: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 9786–9794. https://doi.org/10.1609/aaai.v36i9.21214.

Kaza, S., Yao, L., Bhada-Tata, P., Van Woerden, F., 2018. What a Waste 2.0: a Global Snapshot of Solid Waste Management to 2050. World Bank Publications. https://doi.org/10.1596/978-1-4648-1329-0.

Kiranyaz, S., Avci, O., Abdeljaber, O., Ince, T., Gabbouj, M., Inman, D.J., 2021. 1d convolutional neural networks and applications: a survey. Mech. Syst. Signal Process. 151, 107398. https://doi.org/10.1016/j.ymssp.2020.107398.

Korucu, M., Kucukoglu, I., 2024. Optimization models for long-term planning of municipal solid waste management systems: a review with an emphasis on mass balances. J. Environ. Inform. 43. https://doi.org/10.3808/jei.202400504.

Lebreton, L., Andrady, A., 2019. Future scenarios of global plastic waste generation and disposal. Palgrave. Commun. 5, 1–11. https://doi.org/10.1057/s41599-018-0212-7.

Li, J., Deng, X., Zhan, Z., Yu, L., Tan, K., Lai, K., Zhang, J., 2022. A multipopulation multiobjective ant colony system considering travel and prevention costs for vehicle routing in COVID-19-like epidemics. IEEE Trans. Intell. Transport. Syst. 23 (12), 25062–25076. https://doi.org/10.1109/TITS.2022.3180760.

Liao, S., Li, X., Niu, Y., Xu, Z., Cao, Y., 2024a. Risk control of epidemic in urban cold-chain transportation. Sustain. Cities Soc. 107, 105408. https://doi.org/10.1016/j.scs.2024.105408.

Liao, S., Wu, Y., Ma, K., Niu, Y., 2024b. Ant colony optimization with look-ahead mechanism for dynamic traffic signal control of IoV systems. IEEE Internet Things J. 11, 366–377. https://doi.org/10.1109/JIOT.2023.3286799.

Liu, T., Cao, J., Miatto, A., 2024. Impacts of a municipal solid waste classification policy on carbon emissions: case study of Beijing, China. J. Mater. Cycles Waste Manag. 1–13. https://doi.org/10.1007/s10163-024-01985-9.

Lu, W., Huo, W., Gulina, H., Pan, C., 2022. Development of machine learning multi-city model for municipal solid waste generation prediction. Front. Environ. Sci. Eng. 16, 119. https://doi.org/10.6201007/s11783-022-15516.

Maalouf, A., Mavropoulos, A., 2023. Re-assessing global municipal solid waste generation. Waste Manag. Res. 41, 936–947. https://doi.org/10.1177/0734242X221074116.

Moazzeni, S., Tavana, M., Darmian, S.M., 2022. A dynamic location-arc routing optimization model for electric waste collection vehicles. J. Clean. Prod. 364, 132571. https://doi.org/10.1016/j.jclepro.2022.132571.

Niu, Y., Kong, D., Wen, R., Cao, Z., Xiao, J., 2021a. An improved learn-able evolution model for solving multi-objective vehicle routing problem with stochastic demand. Knowl. Base Syst. 230, 107378. https://doi.org/10.1016/j.knosys.2021.107378.

Niu, Y., Shao, J., Xiao, J., Song, W., Cao, Z., 2022. Multi-objective evolutionary algorithm based on RBF network for solving the stochastic vehicle routing problem. Inf. Sci. 609, 387–410. https://doi.org/10.1016/j.ins.2022.07.087.

Niu, Y., Su, Y., Li, S., Wan, S., Cao, X., 2023. Deep adversarial autoencoder recommendation algorithm based on group influence. Inf. Fusion 100, 101903. https://doi.org/10.1016/j.inffus.2023.101903.

Niu, Y., Xu, C., Liao, S., Zhang, S., Xiao, J., 2024. Multi-objective location-routing optimization based on machine learning for green municipal waste management. Waste. Manage. 181, 157–167. https://doi.org/10.1016/j.wasman.2024.04.001.

Niu, Y., Zhang, Y., Cao, Z., Gao, K., Xiao, J., Song, W., Zhang, F., 2021b. MIMOA: a membrane-inspired multi-objective algorithm for green vehicle routing problem with stochastic demands. Swarm Evol. Comput. 60, 100767. https://doi.org/10.1016/j.swevo.2020.100767.

Pena, D., Dorronsoro, B., Ruiz, P., 2024. Sustainable waste collection optimization using electric vehicles. Sustain. Cities Soc. 105, 105343. https://doi.org/10.1016/j.scs.2024.105343.

Pons-Prats, J., Živojinović, T., Kuljanin, J., 2022. On the understanding of the current status of urban air mobility development and its future prospects: commuting in a flying vehicle as a new paradigm. Transport. Res. E Logist. Transport. Rev. 166, 102868. https://doi.org/10.1016/j.tre.2022.102868.

Singh, S., Chhabra, R., Arora, J., 2023. A systematic review of waste management solutions using machine learning, internet of things and blockchain technologies: state-of-art, methodologies, and challenges. Arch. Comput. Methods Eng. 1–22. https://doi.org/10.1007/s11831-023-10008-z.

Tan, K., Cheong, C., Goh, C., 2007. Solving multiobjective vehicle routing problem with stochastic demand via evolutionary computation. Eur. J. Oper. Res. 177, 813–839. https://doi.org/10.1016/j.ejor.2005.12.029.

United Nations, H, 2022. World Cities Report 2022: Envisaging the Future of Cities. Publisher: UN-Habitat.

Van England, J., Belien, J., De Boeck, L., De Jaeger, S., 2020. Literature review: strategic network optimization models in waste reverse supply chains. Omega 91, 102012. https://doi.org/10.1016/j.omega.2018.12.001.

Wan, S., Nik-Bakht, M., Ng, K.T.W., Tian, X., An, C., Sun, H., Yue, R., 2024. Insights into the urban municipal solid waste generation during the covid-19 pandemic from machine learning analysis. Sustain. Cities Soc. 100, 105044. https://doi.org/10.1016/j.scs.2023.105044.

Wang, L., Wang, S., Zhou, Y., Liu, W., Hou, Y., Zhu, J., Wang, F., 2018. Mapping population density in China between 1990 and 2010 using remote sensing. Remote. Sensing. Environ. 210, 269–281. https://doi.org/10.1016/j.rse.2018.03.007.

Wen, M., Sun, W., Yu, Y., Tang, J., Ikou, K., 2022. An adaptive large neighborhood search for the larger-scale multi depot green vehicle routing problem with time windows. J. Clean. Prod. 374, 133916. https://doi.org/10.1016/j.jclepro.2022.133916.

Yazdani, M., Kabirifar, K., Frimpong, B.E., Shariati, M., Mirmozaffari, M., Boskabadi, A., 2021. Improving construction and demolition waste collection service in an urban area using a simheuristic approach: a case study in Sydney, Australia. J. Clean. Prod. 280, 124138. https://doi.org/10.1016/j.jclepro.2020.124138.

Zambrano-Monserrate, M.A., Ruano, M.A., Ormeño-Candelario, V., 2021. Determinants of municipal solid waste: a global analysis by countries' in-come level. Environ. Sci. Pollut. Control Ser. 28, 62421–62430. https://doi.org/10.1007/s11356-021-15167-9.

Zhang, X., Tian, Y., Cheng, R., Jin, Y., 2014. An efficient approach to non- dominated sorting for evolutionary multiobjective optimization. IEEE Trans. Evol. Comput. 19, 201–213. https://doi.org/10.1016/10.1109/TEVC.2014.2308305.

Zheng, Q., Seto, K.C., Zhou, Y., You, S., Weng, Q., 2023. Nighttime light remote sensing for urban applications: progress, challenges, and prospects. ISPRS J. Photogrammetry Remote Sens. 202, 125–141. https://doi.org/10.1016/j.isprsjprs.2023.05.028.